Temperature and particle motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a drawing or explanation of particles to distinguish average motion from total energy.

SUCCESS CRITERIA - I can...

1. Describe temperature in terms of microscopic motion.
2. Use kelvin in proportional temperature comparisons.
3. Avoid treating a single particle speed as the temperature.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “kelvin” and “particle” to explain your answer. Sentence starters are optional.

kelvin: the absolute temperature scale used in thermal ratios

particle: a small constituent, such as an atom or molecule

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

An object at **350 K** sits in a vacuum facing surroundings at **230 K**. How can it cool, and are absolute-zero surroundings required?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

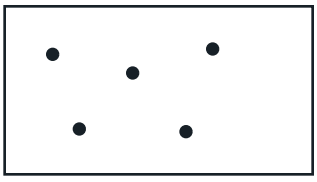
Reference: $K_{avg} \propto T$; $T(K) = T(°C) + \mathbf{273.15}$.
Ideal gas: mean translational kinetic energy per particle scales with absolute temperature. ∝ means proportional to.

Worked reasoning: In a simplified description of a gas, a higher temperature means the particles have greater average kinetic energy from random motion. Individual particles still have a range of speeds.

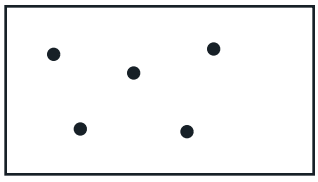
Your turn: Draw two equal-size samples using dots and arrows; **label arrows as particle velocities** and make the warmer sample's average arrow length greater.

Sentence starter: Both ... . They differ in ... because ... .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

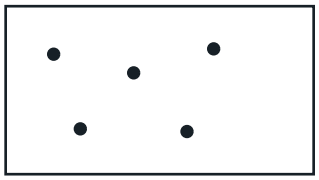
03 Read and explain the diagram or equation

Each box represents a sample of the same ideal gas. One is cooler and one warmer. **Add particle-velocity arrows** with a range of lengths in each box. Make the warmer sample’s average kinetic energy greater, without making every particle identical.

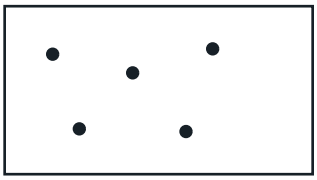
Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

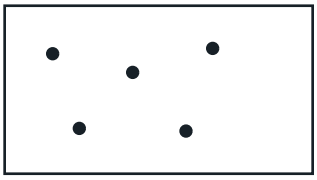
04 Apply the physics

An ideal-gas simulation warms a sample from **300 K** to **600 K**.
1. Use average translational energy = (**3/2**)kT with k = **1.38 × 10⁻²³ J/K**.
2. **Calculate the average energy per** particle at each temperature.
3. **Label the diagram** with the comparison.

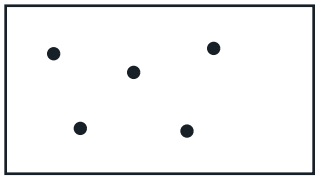
Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student doubles **20°C** to **40°C** and says average particle kinetic energy doubles. **Explain why the temperature scale** matters.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A constant **40 N** forward force acts at steady **6 m/s** for **8 s**. **Find mechanical power** and work.
Use the table to distinguish energy from its transfer rate.

Word help: power = rate of energy transfer

Sentence starter: I found … [unit]. This value means … in this situation.

Case	Energy (J)	Time (s)	Power (W)
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

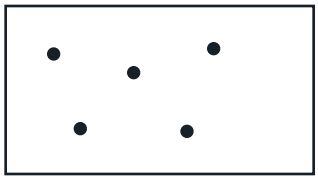
07 Exit check

For a fixed ideal-gas particle mass, average kinetic energy doubles. By what factor does rms speed change?

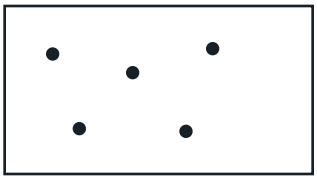
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Sentence starter: I notice … . This supports … because … .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

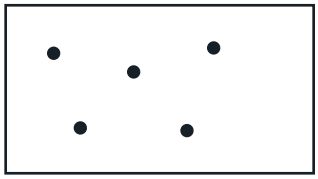
08 STRETCH: test the prediction

For the same ideal gas, typical molecular speed scales with√T in kelvin **0**.
1. **Compare typical speeds** at **300 K** and **1200 K**.
2. Why is a Celsius ratio inappropriate?

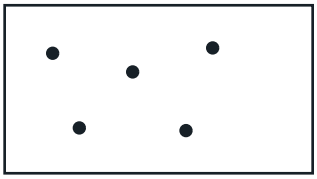
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Sample B



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